We create chemistry

Safety data sheet

BASF Safety data sheet according to Regulation (EC) No. 1907/2006 as amended from time to time.
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Product: **Basolon GL 40 exp**
(ID no. 30735178/SDS_GEN_IE/EN)
Date of print 09.07.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Basolon GL 40 exp

UFI: DQXE-30GV-300E-RDA4

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Performance Chemicals for Oilfield Applications

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF Ireland DAC
Asgard House, 19-20 City Quay
Dublin, D02 K744
Ireland

Telephone: +353 21 451-7100
E-mail address: product-safety-uk-and-ireland@basf.com

1.4. Emergency telephone number

For products classified as hazardous in accordance with CLP:
National Poisons Information Centre, Beaumont Hospital, Dublin 9
Emergency medical information: 8am-10pm (seven days)
Tel.: 01 8092566
International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

Acute Tox. 4 (Inhalation - mist)	H332 Harmful if inhaled.
Skin Corr./Irrit. 2	H315 Causes skin irritation.
Eye Dam./Irrit. 2	H319 Causes serious eye irritation.
Skin Sens. 1	H317 May cause an allergic skin reaction.
Muta. 2	H341 Suspected of causing genetic defects.

Some substances (acids, bases, etc.) are placed on the market in aqueous solutions at various concentrations and, therefore, these solutions require different labelling since the hazards vary at different concentrations. In Annex I entries with Note B have a general designation of the following type: 'nitric acid ...%'. In this case the manufacturer or any other person who markets such a substance in aqueous solution must state the percentage concentration of the solution on the label. Example: nitric acid 45 %. Unless otherwise stated, it is assumed that the percentage concentration is calculated on a weight/weight basis. The use of additional data (e.g. specific gravity, degrees Baumé) or descriptive phrases (e.g. fuming or glacial) is permissible.

STOT SE 3	H335 May cause respiratory irritation.

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Warning

Hazard Statement:

H319	Causes serious eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H341	Suspected of causing genetic defects.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P271	Use only outdoors or in a well-ventilated area.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Hazard determining component(s) for labelling: glyoxal…%

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.
The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical nature

mixture

Regulatory relevant ingredients

glyoxal…%

Content (W/W): >= 39.5 % - <= 40.5 % CAS Number: 107-22-2 EC-Number: 203-474-9 REACH registration number: 01-2119461733-37 INDEX-Number: 605-016-00-7	Acute Tox. 4 (Inhalation - mist) Skin Corr./Irrit. 2 Eye Dam./Irrit. 2 Skin Sens. 1 Muta. 2 STOT SE 3 (irr. to respiratory syst.) H319, H315, H332, H317, H335, H341

ethanediol

	Content (W/W): >= 0 % - <= 2.5 % CAS Number: 107-21-1 EC-Number: 203-473-3 REACH registration number: 01-2119456816-28 INDEX-Number: 603-027-00-1 Substance with EU occupational exposure limit	Acute Tox. 4 (oral) STOT RE (Kidney) 2 H302, H373
Glycollic acid	Content (W/W): >= 0.05 % - <= 0.35 % CAS Number: 79-14-1 EC-Number: 201-180-5	Acute Tox. 4 (oral) Acute Tox. 4 (Inhalation - mist) Skin Corr./Irrit. 1B Eye Dam./Irrit. 1 H314, H302 + H332 EUH071

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

5.2. Special hazards arising from the substance or mixture

Endangering substances: nitrogen oxides, carbon oxides
Advice: The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures

Avoid contact with the skin, eyes and clothing.

6.2. Environmental precautions

Do not discharge into drains/surface waters/groundwater.

6.3. Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulations.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Ensure thorough ventilation of stores and work areas.

7.2. Conditions for safe storage, including any incompatibilities

Suitable materials for containers: Stainless steel 1.4401, Stainless steel 1.4301 (V2), High density polyethylene (HDPE), glass, Low density polyethylene (LDPE)
Unsuitable materials for containers: Paper/Fibreboard
Further information on storage conditions: Protect from air.

Storage stability:
Storage temperature: < 50 °C
Storage duration: 6 Months
May yellow after lengthy storage.
During the storage a product characteristic clouding or crystallin precipitation of the trimeric glyoxal hydrate may occur.
The process is reversible if warmed up mild (max. 40 °C).
From the data on storage duration in this safety data sheet no agreed statement regarding the warrantee of application properties can be deduced.

Protect from temperatures above:50 °C
The packed product must be protected against exceeding the indicated temperature.

7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

107-21-1: ethanediol

Skin Designation (OEL (EU))
The substance can be absorbed through the skin.
STEL value 104 mg/m3 ; 40 ppm (OEL (EU))
indicative
TWA value 52 mg/m3 ; 20 ppm (OEL (EU))
indicative
TWA value 20 mg/m3 (OEL (IE)), vapour
Indicative OELV
TWA value 52 mg/m3 ; 20 ppm (OEL (IE))
Indicative OELV
STEL value 40 mg/m3 ; 104 ppm (OEL (IE))
Ceiling limit value/factor: 15 min
Indicative OELV

Skin Designation (OEL (IE))
The substance can be absorbed through the skin.
STEL value 104 mg/m3 ; 40 ppm (OEL (IE))
Ceiling limit value/factor: 15 min
Indicative OELV
107-22-2: glyoxal…%
TWA value 0.1 mg/m3 (OEL (IE)), Inhalable fraction and vapor
TWA value 0.1 mg/m3 (OEL (IE)), Inhalable fraction and vapor

PNEC

freshwater: 0.319 mg/l

marine water: 0.0319 mg/l

intermittent release: 1.1 mg/l

sediment (freshwater): 0.685 mg/kg

sediment (marine water): 0.0685 mg/kg

soil: 6.3 mg/kg

STP: 4.1 mg/l

8.2. Exposure controls

Appropriate engineering controls
Provide local exhaust ventilation to control vapours/mists.

Personal protective equipment

Respiratory protection:
Wear respiratory protection if ventilation is inadequate. Gas filter for gases/vapours of organic compounds (boiling point <65 °C, f.e. EN 14387 Type AX)

Consider the risk management measures as outlined in the exposure scenario.

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc. Manufacturer's directions for use should be observed because of great diversity of types.

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Avoid contact with the skin, eyes and clothing. When using, do not eat, drink or smoke. Avoid contact with skin and eyes. Remove soiled or soaked clothing immediately. Wash off any contamination that gets onto the skin with plenty of water and soap, skin care.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid	
Colour:	colourless to yellow	
Odour:	aldehyde-like	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
melting range:	-50 - -15 °C	(OECD Guideline 102)
Boiling point:	103.6 °C (1,013.25 hPa)	(Directive 92/69/EEC, A.2)
Flammability:	not flammable	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Flash point:	> 93 °C	(ISO 13736)
Auto-ignition temperature:	285 °C	(DIN 51794)
Thermal decomposition:	No decomposition if correctly stored and handled.	
pH value:	2 - 3 (40 %(m), 20 °C)	
Viscosity, kinematic:	6.6 mm2/s (20 °C)	(OECD 114)
	3.38 mm2/s (40 °C)	(OECD 114)
Viscosity, dynamic:	8.37 mPa.s (20 °C)	(OECD 114)
	4.25 mPa.s (40 °C)	(OECD 114)
Solubility in water:	miscible (20.2 °C)	(internal method)
Partitioning coefficient n-octanol/water (log Kow):	-1.15 (23 °C; pH value: 7)	(Directive 92/69/EEC, A.8)

	-1 (23 °C; pH value: 5)	(Directive 84/449/EEC, A.8)
	-1.62 (23 °C; pH value: 9)	(Directive 84/449/EEC, A.8)
Vapour pressure:	20.2 hPa (20 °C) Information applies to the solvent.	(Directive 92/69/EEC, A.4)
	106.7 hPa (50 °C) Information applies to the solvent.	(Directive 92/69/EEC, A.4)
Relative density:	1.27 (20 °C)	(Directive 92/69/EEC, A.3)
Density:	approx. 1.27 g/cm3 (40 %(m), 20 °C) 1.2514 g/cm3 (50 °C)	(Directive 92/69/EEC, A.3)

Relative vapour density (air):
Water content greater than 10%.

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: Based on the chemical structure there is no indication of explosive properties.

Impact sensitivity:
Based on the chemical structure there is no shock-sensitivity.

Oxidizing properties

Fire promoting properties: Based on its structural properties the product is not classified as oxidizing.

Pyrophoric properties

Self-ignition temperature: Test type: Spontaneous self-ignition at room-temperature.

Based on its structural properties the product is not classified as self-igniting.

Self-heating substances and mixtures

Self heating ability: It is not a substance capable of spontaneous heating.

Substances and mixtures, which emit flammable gases in contact with water

Formation of flammable gases:
Forms no flammable gases in the presence of water.

Corrosion to metals

	5.2 mm/a	(UN Test C.1 (corrosive to metals))
	No corrosive effect on metal.	

Other safety characteristics

pKA:	The substance does not dissociate.	
Adsorption/water - soil:	KOC: 2.1; log KOC: 0.32	(OECD Guideline 121)
Surface tension:	Based on chemical structure, surface activity is not to be expected.	
Molar mass:	58.04 g/mol	
SAPT-Temperature:	Study scientifically not justified.	
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:	No corrosive effect on metal.	
Formation of flammable gases:	Remarks:	Forms no flammable gases in the presence of water.

10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

10.3. Possibility of hazardous reactions

Reacts with strong alkalies. Exothermic reaction.

10.4. Conditions to avoid

Temperature: > 50 °C
See SDS section 7 - Handling and storage.

10.5. Incompatible materials

Substances to avoid:
strong alkalies

10.6. Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:
Of low toxicity after single ingestion. Virtually nontoxic after a single skin contact. Of moderate toxicity after short-term inhalation.

Experimental/calculated data:
LD50 rat (oral): 3,300 mg/kg (OECD Guideline 401)

LC50 rat (by inhalation): 2.44 mg/l 4 h (OECD Guideline 403)
An aerosol was tested.

LD50 rat (dermal): > 2,000 mg/kg (OECD Guideline 402)
Limit concentration test only (LIMIT test).

Irritation

Assessment of irritating effects:
Eye contact causes irritation. Skin contact causes irritation.

Experimental/calculated data:
Skin corrosion/irritation
rabbit: Irritant. (OECD Guideline 404)

Serious eye damage/irritation
rabbit: Irritant. (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Caused skin sensitization in animal studies. Caused sensitization in humans.

Experimental/calculated data:
Guinea pig maximization test guinea pig: skin sensitizing (OECD Guideline 406)

Human Maximization Test human: skin sensitizing (Patch test)
Literature data.

Germ cell mutagenicity

Assessment of mutagenicity:
The substance was mutagenic in various test systems with microorganisms and cell cultures; however, these results could not be confirmed in tests with mammals.

Carcinogenicity

Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation.

Aspiration hazard

No aspiration hazard expected.

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. Based on long-term (chronic) toxicity study data, the product is very likely not harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:
LC50 (96 h) > 186 - < 272 mg/l, Leuciscus idus (DIN 38412 Part 15, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Aquatic invertebrates:
EC50 (48 h) 161 mg/l, Daphnia magna (Directive 79/831/EEC, static)
Nominal concentration. The ecological data given are those of the active ingredient.

LC50 (96 h) 76.0 mg/l, Americamysis bahia (other, semistatic)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Aquatic plants:
EC50 (72 h) > 40 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

EC10 (72 h) > 10 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

EC50 (72 h) 347.1 mg/l (growth rate), Skeletonema costatum (OECD Guideline 201)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

No observed effect concentration (72 h) 118.4 mg/l (growth rate), Skeletonema costatum (OECD Guideline 201)

The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Microorganisms/Effect on activated sludge:
EC10 (16 h) 22.8 mg/l, Pseudomonas putida (DIN 38412 Part 8, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

EC50 (0.5 h) > 400 mg/l, activated sludge (OECD Guideline 209, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Chronic toxicity to fish:
No observed effect concentration (34 d) 112 mg/l, Pimephales promelas (OPP 72-4 (EPA-Guideline), Flow through.)
The ecological data given are those of the active ingredient. The statement of the toxic effect relates to the analytically determined concentration.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) 3.19 mg/l, Daphnia magna (OECD Guideline 211, semistatic)
The statement of the toxic effect relates to the analytically determined concentration. The ecological data given are those of the active ingredient.

Assessment of terrestrial toxicity:
Toxic effects have been observed in studies with soil living organisms. Toxic effects have been observed in studies with terrestric plants.

Soil living organisms:
LC50 (14 d) > 135 mg/kg, Eisenia foetida (OECD Guideline 207, artificial soil)
The ecological data given are those of the active ingredient. The details of the toxic effect relate to the nominal concentration.

EC50 (28 d) > 1,054 mg/kg, soil dwelling microorganisms (OECD 217, natural soil)
The details of the toxic effect relate to the nominal concentration. The data have been calculated from values for a preparation with a lower substance concentration. The ecological data given are those of the active ingredient.

EC50 (28 d) > 1,054 mg/kg, soil dwelling microorganisms (OECD 216, natural soil)
The details of the toxic effect relate to the nominal concentration. The data have been calculated from values for a preparation with a lower substance concentration. The ecological data given are those of the active ingredient.

Terrestrial plants:
No observed effect concentration (21 d) 314 mg/kg, Brassica napus (OECD Guideline 208)
The ecological data given are those of the active ingredient. The data have been deduced from values for a preparation or mixture with a lower substance concentration.

Other terrestrial non-mammals:
No data available.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Readily biodegradable (according to OECD criteria).

Elimination information:
90 - 100 % DOC reduction (19 d) (OECD 301 A (new version)) (aerobic, activated sludge, domestic, non-adapted) Readily biodegradable (according to OECD criteria).

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.

Information on Stability in Water (Hydrolysis):
The product has not been tested. The statement has been derived from the structure of the product.

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Bioaccumulation potential:

Bioconcentration factor(BCF): 3.2, Fish (calculated)

12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

12.6. Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

12.7. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

12.8. Additional information

Sum parameter

Chemical oxygen demand (COD): 350 mg/g

Biochemical oxygen demand (BOD) Incubation period5 d: 175 mg/g

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Incinerate in suitable incineration plant, observing local authority regulations.

A waste code in accordance with the European waste catalog (EWC) cannot be specified, due to dependence on the usage.
The waste code in accordance with the European waste catalog (EWC) must be specified in cooperation with disposal agency/manufacturer/authorities.

Contaminated packaging:
Uncontaminated packaging can be recycled.

SECTION 14: Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable

Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel

UN number or ID number:	ID9006
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (GLYOXAL)
Transport hazard class(es):	9, N3
Packing group:	Not applicable
Environmental hazards:	yes
Type of inland waterway vessel:	N
Cargo tank design:	4
Cargo tank type:	2

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Regulation:	IBC-Code
Product name:	Glyoxal solution (40% or less)
Pollution category:	Y
Ship Type:	3

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006
Number on List: 29

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Acute Tox. 4 (Inhalation - mist)
Skin Corr./Irrit. 2
Eye Dam./Irrit. 2B
Aquatic Acute 3
Acute Tox. 5 (oral)
STOT SE 3 (irritating to respiratory system)
Skin Sens. 1

Acute Tox.	Acute toxicity
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Skin Sens.	Skin sensitization
Muta.	Germ cell mutagenicity
STOT SE	Specific target organ toxicity — single exposure
STOT RE	Specific target organ toxicity — repeated exposure
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H341	Suspected of causing genetic defects.
H302	Harmful if swallowed.
H373	May cause damage to organs (Kidney) through prolonged or repeated exposure.
H314	Causes severe skin burns and eye damage.
H302 + H332	Harmful if swallowed or if inhaled
EUH071	Corrosive to the respiratory tract.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Annex: Exposure Scenarios

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1. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures
SU3; ERC2; PROC3, PROC8b, PROC9

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ESVOC SpERC 4.10a.v1: ESVOC SpERC 4.10a.v1
Operational conditions	
Annual amount used in the EU	100,000 kg
Minimum emission days per year	300
Emission factor air	2.5 %
Emission factor water	0.5 %
Emission factor soil	0.01 %
Receive Surf. Water (Flow Rate).	18,000 m3/d
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Air treatment measures considered suitable are, e.g.	Wet scrubber - for dusts, Filtration, Waste gas treatment by thermal oxidation, Adsorption

Wastewater treatment measures considered suitable are, e.g.	Acclimated biological treatment, Distillation
Type of STP	Municipal STP
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, Environment
Risk Characterization Ratio (RCR)	0.056667
	Risk from environmental exposure is driven by marine sediment.
Maximum amount of safe use	5,882.3 kg/d
Risk from environmental exposure is driven by marine sediment.	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic

Exposure estimate	0.0329 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.004987
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	1.7411 mg/m³
Risk Characterization Ratio (RCR)	0.588204
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.
Risk Management Measures	
Local exhaust ventilation	Effectiveness: 95 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	1.0971 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.166234
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic

Exposure estimate	1.2091 mg/m³
Risk Characterization Ratio (RCR)	0.408475
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.
Risk Management Measures	
Local exhaust ventilation	Effectiveness: 90 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	0.3291 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.04987
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	2.0313 mg/m³
Risk Characterization Ratio (RCR)	0.686238
Guidance to Downstream Users	

For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)

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2. Short title of exposure scenario

Use as Reactive process agent
SU3; SU3, SU6b; ERC6b; PROC7, PROC10, PROC13

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC6b: Use of reactive processing aid at industrial site (no inclusion into or onto article)
Operational conditions	
Annual amount per site	500,000 kg
Minimum emission days per year	300
Emission factor air	0.002 %
Emission factor water	0.1 %
Emission factor soil	0.1 %
Receive Surf. Water (Flow Rate).	18,000 m3/d
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Wastewater treatment measures considered suitable are, e.g.	Stripping, Thermal Treatment - Distillation / Rectification
Type of STP	Municipal STP
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Waste-Related Measures	
Prescribed disposal method	waste combuster
Effectiveness	0 %
Exposure estimate and reference to its source	
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, Environment
Risk Characterization Ratio (RCR)	0.040678
	Risk from environmental exposure is driven by freshwater sediment.
Maximum amount of safe use	40,971.8 kg/d
Risk from environmental exposure is driven by freshwater sediment.	

Contributing exposure scenario	
Use descriptors covered	PROC7: Industrial spraying Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 10 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Ensure that the task is being carried out outside the breathing zone of a worker (distance head-product greater than 1m).	
Ensure that the worker is in a seperated (control) room with independent air supply Ensure that a spraying booth is used. Ensure containment of the emission source	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v4.2, RISKOFDERM v2.1
	Worker - dermal, long-term - systemic
Exposure estimate	0.26 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.039394
Assessment method	EASY TRA v4.2, Stoffenmanager v5.6
	Worker - inhalation, long-term - systemic
Exposure estimate	0.0001 mg/m³
Risk Characterization Ratio (RCR)	0.000001

Contributing exposure scenario	
Use descriptors covered	PROC10: Roller application or brushing Use domain: industrial
Operational conditions	

Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Local exhaust ventilation	Effectiveness: 90 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	2.1943 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.332468
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	1.4509 mg/m³
Risk Characterization Ratio (RCR)	0.49017
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

Contributing exposure scenario	
Use descriptors covered	PROC13: Treatment of articles by dipping and pouring. Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %

Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Local exhaust ventilation	Effectiveness: 90 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	1.0971 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.166234
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	1.4509 mg/m³
Risk Characterization Ratio (RCR)	0.49017
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

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3. Short title of exposure scenario

Use in Adhesives, Use in Coatings
SU3; SU3, SU18; ERC6b; PROC1, PROC3, PROC4, PROC5

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC6d: Use of reactive process regulators in

	polymerisation processes at industrial site (inclusion or not into/onto article)
Operational conditions	
Annual amount used in the EU	40,000 kg
Minimum emission days per year	20
Emission factor air	35 %
Emission factor water	0.005 %
Emission factor soil	0.025 %
Receive Surf. Water (Flow Rate).	18,000 m3/d
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Type of STP	Municipal STP
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, Environment
Risk Characterization Ratio (RCR)	0.02185
	Risk from environmental exposure is driven by marine sediment.
Maximum amount of safe use	91,534.4 kg/d
Risk from environmental exposure is driven by marine sediment.	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.

Risk Management Measures	
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	0.0027 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.000416
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	0.0097 mg/m³
Risk Characterization Ratio (RCR)	0.003268
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Use suitable chemically resistant	Effectiveness: 80 %

gloves.	
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	0.0329 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.004987
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	1.7411 mg/m³
Risk Characterization Ratio (RCR)	0.588204
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

Contributing exposure scenario	
Use descriptors covered	PROC4: Chemical production where opportunity for exposure arises Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Local exhaust ventilation	Effectiveness: 90 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision	

in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	0.5486 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.083117
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	1.9345 mg/m³
Risk Characterization Ratio (RCR)	0.65356
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.
Risk Management Measures	
Local exhaust ventilation	Effectiveness: 90 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	

Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	0.6583 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.09974
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	2.0313 mg/m³
Risk Characterization Ratio (RCR)	0.686238
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

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4. Short title of exposure scenario

Use in gas treatment, Use in Oil and Gas field drilling and production operations
SU3; SU2, SU3; ERC6b; PROC1, PROC2, PROC3; PC2

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ESVOC SpERC 2.2.v1: ESVOC SpERC 2.2.v1
Operational conditions	
Annual amount used in the EU	20,000 kg
Minimum emission days per year	300
Emission factor air	0.01 %
Emission factor water	0.001 %
Emission factor soil	0 %
Receive Surf. Water (Flow Rate).	18,000 m3/d
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Air treatment measures considered suitable are, e.g.	Adsorption
Wastewater treatment measures considered suitable are, e.g.	Acclimated biological treatment
Type of STP	Municipal STP

Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, Environment
Risk Characterization Ratio (RCR)	0.019627
	Risk from environmental exposure is driven by marine sediment.
Maximum amount of safe use	6.8 kg/d
Risk from environmental exposure is driven by marine sediment.	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.
Risk Management Measures	
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	0.0027 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.000416
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	0.0097 mg/m³

Risk Characterization Ratio (RCR)	0.003268
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	0.1097 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.016623
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	1.4509 mg/m³
Risk Characterization Ratio (RCR)	0.49017
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see	

exposure estimates)

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Use domain: industrial
Operational conditions	
Concentration of the substance	glyoxal…% Content: >= 0 % - <= 40 %
Physical state	liquid
Vapour pressure of the substance during use	2020.000076 Pa
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Assumes activities are at ambient temperature.
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Use suitable chemically resistant gloves.	Effectiveness: 80 %
Avoid frequent and direct contact with substance. Ensure minimization of manual phases Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - dermal, long-term - systemic
Exposure estimate	0.0329 mg/kg bw/day
Risk Characterization Ratio (RCR)	0.004987
Assessment method	EASY TRA v5.2, ECETOC TRA v3.0, worker, modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	1.7411 mg/m³
Risk Characterization Ratio (RCR)	0.588204
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra Please note that a modified version has been used (see exposure estimates)	

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